Precision Tissue Mapping from Spatial Transcriptomics Using Graph Learning

A neighborhood-aware framework for spatially resolved molecular pathology

Akshatha Arunkumar
Student Research Institute (SRI) – Summer 2026 Cohort
Harvard Undergraduate OpenBio Laboratory

Research Proposal for Mentor Review | *June 15, 2026*

The problem. Spatial transcriptomics now measures gene expression *while preserving each cell's location in a tissue*, letting us ask not only *which* genes change in disease but *where* in the tissue the change happens – the question at the heart of precision oncology and neuropathology [1]. But before any spatial slide can be interpreted, every measured spot must be assigned to its **tissue domain** (a cortical layer, a tumor region, an immune niche). Today this is done by slow, subjective manual annotation, or by expression-only clustering that ignores location and produces noisy, spatially incoherent maps. **This project builds an automated, label-efficient model for that mapping step and shows it must be *spatially aware* to work.**

Preliminary result – already obtained on real human & mouse tissue

On the gold-standard **LIBD DLPFC** benchmark (12 Visium sections, 47,329 expert-annotated spots, 7 cortical domains), our spatial GNN reaches **macro-F1** 0.75 / **ARI** 0.65 under the hard *cross-section* split (whole tissues held out, 3 seeds), versus **XGBoost** 0.52 and **MLP** 0.49 on *identical* expression features – a +0.24 **macro-F1** win. The result is **architecture-robust** (GraphSAGE and a graph-attention GAT encoder both ≈ 0.75 F1) and **transfers to a second technology**: on single-cell *osmFISH* mouse cortex (33 genes) the GNN hits **macro-F1** 0.96 vs 0.54 (XGBoost) / 0.63 (MLP). A **graph-removal falsification test on both tissues** collapses the identical GNN back onto the blind MLP, proving the gain is *spatial*, not just "a bigger model." The proposal below formalizes, stress-tests, and extends these results toward tumor-microenvironment mapping.

1 Background and Significance

Why spatial context is clinically important. Tissue is not a bag of cells – its function is organized in space. In the cerebral cortex, six layers (L1–L6) above the white matter each carry distinct cell types and gene programs; in a tumor, malignant cells, stroma, blood vessels, and infiltrating immune cells form spatially structured *niches*. Crucially, disease changes are *spatially localized*: tau pathology in Alzheimer's follows a laminar gradient, schizophrenia shows layer-specific cortical deficits, and a tumor's response to immunotherapy depends on *where* immune cells sit relative to the malignant front. Standard bulk RNA sequencing averages this away. Spatial transcriptomics (10x Visium, MERFISH, Xenium, CosMx, Stereo-seq) preserves it, and a 2024 review in *J. Cancer Res. Clin. Oncol.* argues this spatial information is poised to reshape precision oncology by capturing tumor heterogeneity, metastasis, prognosis, and immunotherapy responsiveness that bulk profiling misses [1].

Why it matters	Consequence
Tissue function is spatially organized	Disease changes are region/layer-specific, not diffuse.
Spatial omics preserves location	We can finally ask *where* pathology starts.
But spots must be assigned to a domain first	Domain mapping is the rate-limiting step for *every* downstream spatial analysis.

Table 1: Accurate spatial-domain mapping is the enabling step that turns a spatial-transcriptomics slide into interpretable, region-resolved biology – in brain disease and in cancer alike.

2 The Computational Gap

Domain mapping is hard for two complementary reasons, and current practice addresses neither well:

Gap 1 – Manual annotation does not scale. Expert annotation of a single section is slow and subjective; a disease cohort spans dozens of sections across many donors, where hand-labelling becomes the bottleneck.

Gap 2 – Per-spot expression is weak and structure-blind. A single spot's expression is sparse and dropout-corrupted, so models that classify each spot *in isolation* (XGBoost, MLP) hit an accuracy ceiling and yield spatially fragmented domains.

The key observation we exploit is simple but decisive: **a spot's domain is a property of its spatial neighbourhood.** Adjacent spots overwhelmingly share a domain, so the signal missing from one noisy spot is present in the tissue around it (Fig. 1). A model that can *see the neighbourhood* should therefore beat one that cannot – and the size of that gap measures exactly how much spatial structure is worth.

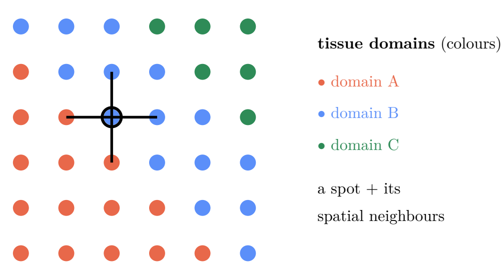

Figure 1: Domains are spatially contiguous, so a spot's label agrees with its neighbours'. A single (noisy) spot may be ambiguous, but its neighbourhood reveals its domain – the signal a spatial model can use and a per-spot model cannot.

3 Central Hypothesis and Specific Aims

Central hypothesis. A spot's tissue domain is encoded not only in its own gene expression but in its *spatial neighbourhood*. Therefore a graph neural network that performs tissue-neighbour message passing will map domains more accurately than XGBoost and MLP models trained on the identical expression features – and removing the spatial graph will erase the advantage.

Specific Aims

Aim 1 – Build & benchmark. Develop a spatial GNN for tissue-domain mapping and benchmark it head-to-head against tuned XGBoost and an MLP on identical features, under leakage-safe splits.
Aim 2 – Prove the advantage is spatial. Run a graph-removal falsification test (intact vs. shuffled vs. empty graph) on two tissues / technologies; show the GNN collapses to the MLP when structure is destroyed.
Aim 3 – Ground it biologically + extend to cancer. Recover known domain marker genes, surface candidate annotation errors, and pilot the same pipeline on tumor spatial transcriptomics to map microenvironment niches.

4 Relationship to Prior Work (what is and isn't new)

GNN-based spatial-domain identification is an *established* field: SpaGCN [6], STAGATE, and GraphST [5] all apply graph neural networks to spatial-omics data and are standard baselines on the DLPFC benchmark. **We therefore do not claim a new architecture.** Those methods are predominantly *unsupervised* (graph autoencoders + clustering, scored by ARI against manual labels) and are compared mainly against other clustering methods. Our contribution is a complementary and, in this field, under-asked *question*: **how much does spatial structure actually add over strong non-graph models given the identical features?** We answer it with (i) a *supervised*, leakage-safe *cross-section* protocol (train on whole sections, predict held-out ones – the disease-cohort regime); (ii) *tuned XGBoost and MLP* baselines on the same features, which spatial-domain papers rarely report; (iii) a *graph-removal falsification test* that isolates the spatial signal from model capacity; and (iv) replication across *two technologies* (Visium, osmFISH). The deliverable is a quantified, honestly bounded estimate of the value of tissue structure – a rigorous benchmark and analysis, not a state-of-the-art or novel-method claim.

5 Datasets

Dataset	Content	Role
LIBD DLPFC (10x Visium) [2, 3]	12 sections, 47,329 spots, expert manual layer labels (L1–L6 + WM)	Primary ground-truth benchmark (cross-section)
osmFISH mouse cortex [4]	single-cell resolution, 33 genes, $\sim$4,839 cells, region labels	Second technology / generalization
Controlled synthetic tissue	domains defined purely by neighbourhood; per-spot signal weak by construction	Mechanism isolation / sanity check
Tumor ST (Visium/Xenium, stretch)	public solid-tumor sections with niche annotations	Clinical extension (Aim 3)

Table 2: All data are public and de-identified. LIBD DLPFC is the field-standard ground truth for spatial-domain methods; osmFISH tests a different assay with far fewer genes, where per-cell features are weakest.

6 Methodology

Tissue → spatial graph. Each section becomes a graph: nodes are spots/cells, node features are preprocessed expression (normalize → log1p → highly-variable genes → PCA), and each node is connected to its k nearest *spatial* neighbours. *2D coordinates are used only to build the graph, never as model features*, so any GNN benefit is attributable to message passing over tissue structure, not to memorizing absolute position.

6.1 Model: a spatial graph neural network

The GNN classifies each spot by pooling its neighbourhood over L rounds of message passing. With a GraphSAGE encoder,

$$h_v^{(l)} = \sigma\Big(W^{(l)}\big[\,h_v^{(l-1)} \,\big\|\, \mathrm{AGG}_{u\in\mathcal{N}(v)}\, h_u^{(l-1)}\,\big]\Big), \qquad \hat{y}_v = \mathrm{softmax}\big(W_o\, h_v^{(L)}\big), \tag{1}$$

trained with cross-entropy on annotated spots. Aggregation denoises the sparse per-spot signal – exactly the property Gap 2 lacks. We compare **GraphSAGE**, **GAT** (graph attention, which learns per-neighbour weights), and GCN encoders; all use PyTorch Geometric.

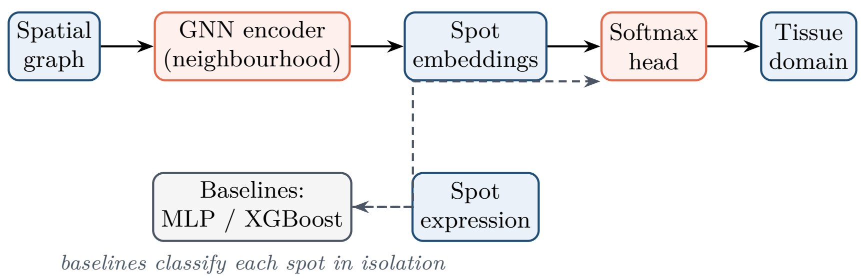

Figure 2: Pipeline. The GNN (orange) pools each spot's spatial neighbourhood; the non-spatial baselines (gray) classify each spot alone from the *same* features. The only difference between them is access to the spatial graph.

6.2 Baselines and leakage-safe evaluation

We benchmark against **tuned XGBoost** and an **MLP** on the same expression features, so the comparison isolates the value of tissue structure. **Splits:** (i) *cross-section* – whole sections held out for test (layer layouts differ per section, so absolute position cannot transfer and only neighbourhood-relative reasoning generalizes; this is the disease-cohort regime and our headline test); (ii) *within-section* – contiguous spatial blocks held out; (iii) *stratified* – standard transductive node split for single-section data (osmFISH). In the cross-section disjoint-union graph, train and test sections are separate connected components, so message passing cannot leak labels across the split. **Metrics:** accuracy, macro-F1 (domains are imbalanced), and adjusted Rand index (ARI, the field-standard domain metric), reported as mean ± std over seeds.

7 Preliminary Results

A working pilot already supports every claim above.

8 Rigor: Falsification and Biological Interpretation

Macro-F1 / ARI	XGBoost	MLP	SAGE	GAT
DLPFC, *cross-section* (3 seeds)	0.52 / 0.37	0.49 / 0.38	**0.75 / 0.65**	**0.74 / 0.66**
osmFISH, *within-section* (3 seeds)	0.54 / 0.44	0.63 / 0.51	**0.96 / 0.95**	**0.97 / 0.95**

Table 3: Spatial GNNs beat strong non-spatial baselines on the *same* features, on two tissues and two technologies. The osmFISH gap is even larger because its 33-gene per-cell signal is weakest, so the neighbourhood matters most. (osmFISH is a single section, so this is a transductive within-section split – an easier regime than DLPFC's cross-section test, and reported as such.)

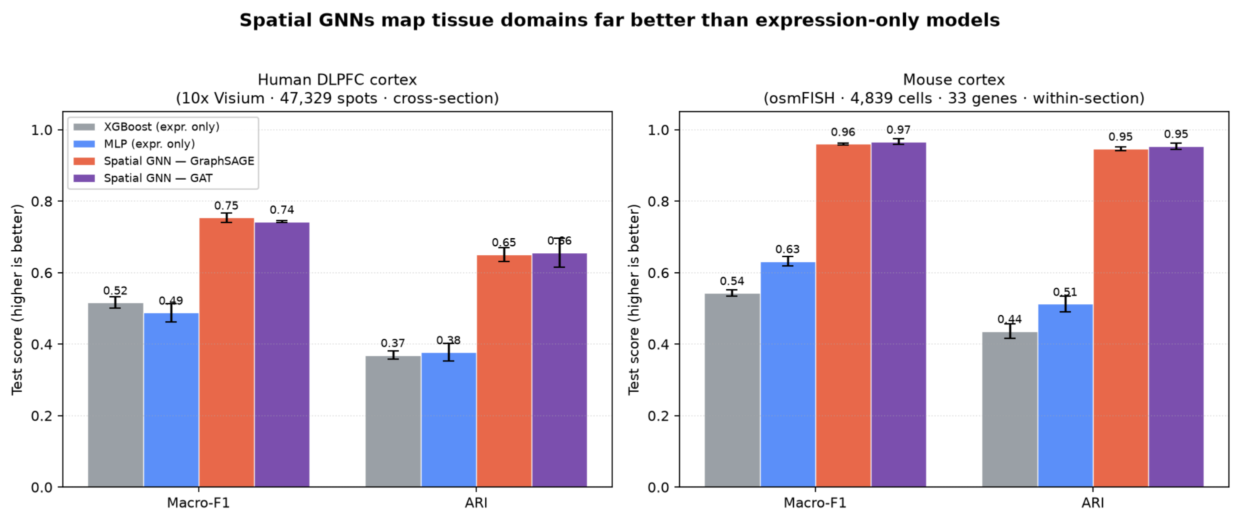

Figure 3: Headline comparison. On both human DLPFC (Visium) and mouse cortex (osmFISH), the spatial GNN (GraphSAGE / GAT) substantially outperforms XGBoost and MLP, which see the same expression features but not the tissue graph.

8.1 Is the spatial structure really doing the work?

We run the *identical* GNN on three graphs. If the gain is genuinely spatial, destroying tissue structure must erase it – and it does, on both tissues (Fig. 4, Table 4).

8.2 From accuracy to biology

For each predicted domain we will (i) extract the genes driving predictions and confirm recovery of **known marker genes** (cortical-layer markers for DLPFC), and (ii) flag spots where the model confidently disagrees with the manual label – candidate *annotation errors* or genuine micro-domains for expert review. This turns an accuracy number into a hypothesis a biologist can act on.

9 Clinical Translation: Toward Precision Pathology

The cortical-layer benchmark is the rigorous *proof of concept*; the same capability – accurate, automated, spatially-coherent domain mapping – is what spatial transcriptomics needs to reach the clinic, especially in oncology [1].

GNN macro-F1	Intact graph	Shuffled edges	Empty graph	MLP ref.
DLPFC (cross-section)	**0.75**	0.49	0.49	0.48
osmFISH (within-section)	**0.95**	0.58	0.61	0.63

Table 4: Falsification test. Replacing the true spatial kNN graph with degree-matched random edges or no edges collapses the GNN onto the blind MLP – on both tissues. The advantage is the *graph*, not the network capacity.

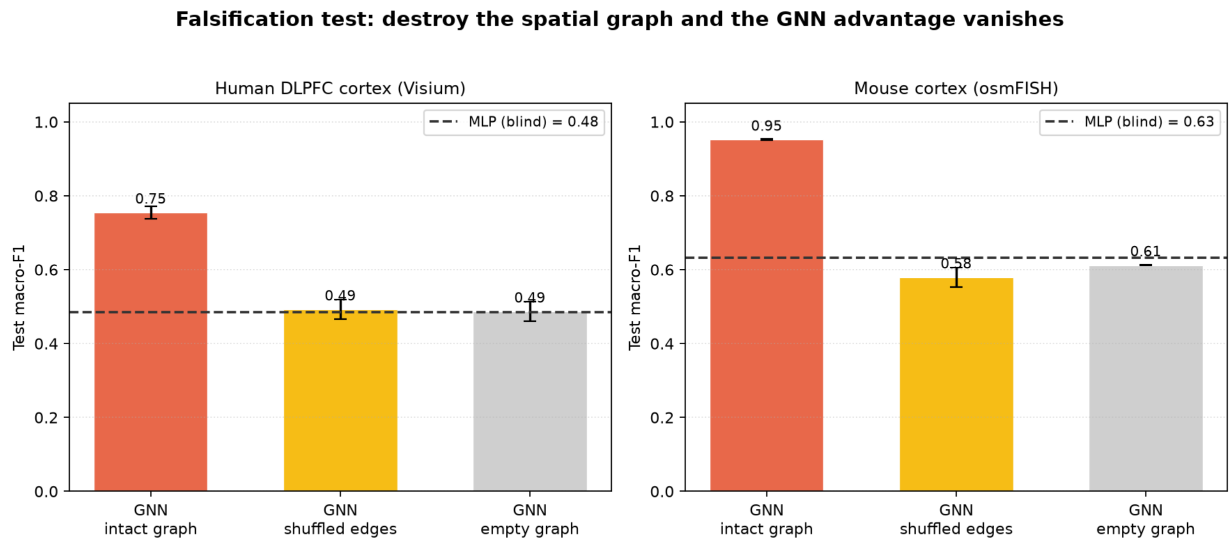

Figure 4: The same falsification test, visualized on both tissues. Only the intact spatial graph clears the MLP reference line; shuffled/empty graphs fall back to it.

10 Alignment with Strong-Research Criteria

The most competitive SRI original-research projects share a recurring pattern – a real biomedical problem, public data, a clear baseline comparison, a causal check, interpretability, and a translational story. This project is deliberately designed to hit each element (Table 6).

11 Eight-Week Timeline

The plan fits a single eight-week cohort, and the first six weeks are already de-risked by the pilot results above (Table 7).

12 Reproducibility and Ethics

Implementation uses open tools (PyTorch Geometric for the GNN; scanpy/anndata for spatial-omics I/O; XGBoost). All code, configs, random seeds, and a data manifest are released; every reported number is multi-seed with standard deviations. No human subjects are involved – all inputs are public, de-identified post-mortem or model-organism tissue. Predictions are research-only and hypothesis-generating and do not constitute clinical guidance.

References

[1] Cilento MA, Sweeney CJ, Butler LM. *Spatial transcriptomics in cancer research and potential clinical impact: a narrative review.* Journal of Cancer Research and Clinical Oncology, 150(6):296, 2024.

Application	How domain mapping helps
Tumor microenvironment	Separate tumor core, invasive front, stroma, and immune niches to study heterogeneity and metastasis.
Immunotherapy response	Localize where immune cells sit relative to malignant cells – a spatial correlate of response.
Neuro-/neurodegenerative disease	Pinpoint which cortical layers carry early molecular pathology (Alzheimer's, schizophrenia).
Pathologist support	Annotate complex molecular slides faster and more reproducibly than manual labelling.

Table 5: The brain benchmark proves the method; precision oncology and pathology are where it pays off. All predictions are research-only and hypothesis-generating.

Criterion	Commitment
Real, sizable public data	LIBD DLPFC: 12 sections, 47k annotated spots + a second technology (osmFISH).
Clear medical motivation	Spatially-resolved localization of disease pathology; precision oncology bridge.
Rigorous evaluation	Tuned XGBoost + MLP baselines; leakage-safe splits; multi-seed macro-F1 + ARI.
Causal / falsification test	Graph-removal ablation on two tissues collapses the GNN to the MLP.
Interpretability → discovery	Recover marker genes; surface candidate annotation errors.
Honesty clause	Any narrowing of the gap (e.g. easier transductive splits) is reported as-is.

Table 6: The project is built around the elements of strong, defensible research.

doi:10.1007/s00432-024-05816-0. PMID: 38850363.

[2] Maynard KR, Collado-Torres L, et al. *Transcriptome- scale spatial gene expression in the human dorsolateral prefrontal cortex*. Nature Neuroscience, 24:425–436, 2021.

[3] Pardo B, et al. *spatialLIBD: an R/Bioconductor package to visualize spatially-resolved transcriptomics data*. BMC Genomics, 23:434, 2022.

[4] Codeluppi S, et al. *Spatial organization of the somatosensory cortex revealed by osmFISH*. Nature Methods, 15:932–935, 2018.

[5] Long Y, et al. *Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST*. Nature Communications, 14:1155, 2023.

[6] Hu J, et al. *SpaGCN: integrating gene expression, spatial location and histology to identify spatial domains using graph convolutional networks*. Nature Methods, 18:1342–1351, 2021.

[7] Hamilton WL, Ying R, Leskovec J. *Inductive representation learning on large graphs (GraphSAGE)*. NeurIPS, 2017.

[8] Veličković P, et al. *Graph Attention Networks*. ICLR, 2018.

Weeks	Milestones
1–2	Data pipeline (DLPFC + osmFISH to AnnData; spatial-graph builder); synthetic sanity check.
3–4	Spatial GNN (SAGE/GAT/GCN) + tuned XGBoost + MLP; shared leakage-safe trainer.
5	Cross-section, within-section, stratified experiments; multi-seed macro-F1/ARI.
6	Falsification ablations (graph removal, depth, neighbourhood size) on both tissues.
7	Biological interpretation (marker recovery, annotation-error flags); tumor-ST pilot.
8	Figures, tables, manuscript draft; reproducibility package.

Table 7: Schedule for the eight-week SRI cohort. Weeks 1–6 are already de-risked by the pilot results above.